

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 27-305-1
Uninterruptible Power Systems (UPS)	Date: 11-02-2025

EDMS: 27-305-1 SPECIFICATION

FOR Uninterruptible Power Systems (UPS)

Issue: -2025/ Rev- 1

- This revision contains option items that must be selected by EDC before bidding.

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1. SCOPE

This specification covers the minimum technical requirements for design, engineering, manufacture, inspection, testing and performance movable, stationary and fixed electronic uninterruptible power systems (UPS) intended to be used in the Egyptian Electricity Distribution Company premises.

2. APPLICABLE CODES AND STANDARDS

Unless otherwise specified in this specification, the UPS shall comply with the latest edition of IEC standards and shall be designed, manufactured, and tested in accordance with applicable IEC standards or approved equivalents, including but not limited to those listed in Table 1.

Table 1: Applicable Codes and Standards

Standard No.	Description
IEC 62040-1	Uninterruptible power systems (UPS) - Part 1: Safety requirements
IEC 62040-2	Uninterruptible power systems (UPS) - Part 2: Electromagnetic compatibility (EMC) requirements
IEC 62040-3	Uninterruptible power systems (UPS) - Part 3: Method of specifying the performance and test requirements
IEC 61643-1	Low-voltage surge protective devices - Part 1: Surge protective devices connected to low-voltage power distribution systems - Requirements and tests
IEC 60896-21	Stationary lead-acid batteries - Part 21: Valve regulated types - Methods of test
IEC 60896-22	Stationary lead-acid batteries - Part 22: Valve regulated types - Requirements
IEC 60364-5-52	Low-voltage electrical installations - Part 5-52: Selection and erection of electrical equipment - Wiring systems
IEC 62485-2	Safety requirements for secondary batteries and battery installations - Part 2: Stationary batteries.
IEEE 1491	IEEE Guide for Selection and Use of Battery Monitoring Equipment in Stationary Applications
IEEE 1184	IEEE Guide for Batteries for Uninterruptible Power Supply Systems

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3. ENVIRONMENTAL CONDITIONS

The performance of the required UPS should be guaranteed under the environmental conditions given in Table (2), any differences in the guaranteed performance should be clearly set out in the offer.

Table 2: Environmental Conditions

Minimum ambient temperature	0°C
Maximum ambient temperature	40°C (50 °C as option).
Storage temperature	-5 to + 55 °C
Maximum relative humidity	95 % non-condensing.
Protection degree	IP: 20
Acoustic noise	70 dB(A) maximum at 1 meter
Maximum altitude	1000 m

4. SYSTEM DESCRIPTION

4.1.General

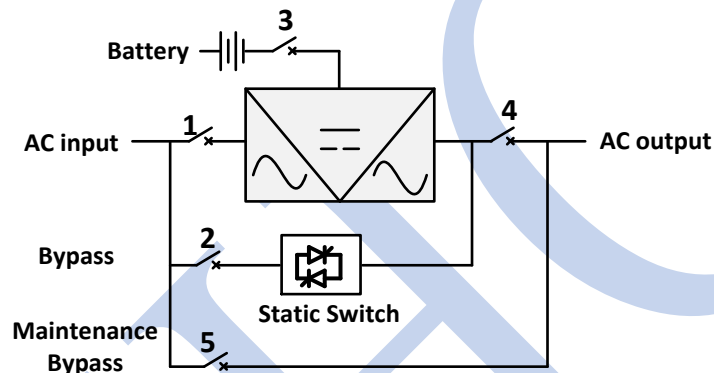
- The UPS shall employ a double-conversion topology and meet the performance requirements of VFI-SS-11 as defined in IEC 62040.
- The UPS shall be a three-phase, four-wire system with a nominal input voltage of 400V.
- The system shall comprise a rectifier/charger, inverter, static bypass switch, manual maintenance bypass, and a sealed, maintenance-free batteries installed in a separate rack.
- The UPS shall ensure a transfer time of less than 5 milliseconds during switching operations.
- The UPS system must comply with IEC 62040-1 for general and safety requirements.
- The UPS system will be floor-mounted in a sturdy metal enclosure.
- All active electronic components shall be solid-state and hermetically sealed. All relays shall be dust-tight.
- All power semiconductors in the module shall be protected by fast-acting fuses to prevent cascading failures in the event of a single component failure. Any fuse blown shall initiate a blown fuse indicator on the control panel.
- Fast-acting current limiting devices shall be employed to protect against failures of solid-state devices.

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4.2. UPS Configuration

4.2.1. Single Unit Normal UPS:

A single UPS with bypass is a single unit equipped with a bypass switch to ensure uninterrupted power supply to the load. Additionally, it includes a maintenance bypass switch to facilitate UPS maintenance without interrupting power to the load.



4.2.2. Parallel Unit Normal UPS:

A parallel UPS with distributed bypass consists of two or more UPS units with their AC outputs connected to a common AC output bus. The batteries bank(s) in a parallel UPS can be either dedicated to individual UPS units or shared among multiple units.

This configuration features paralleled UPS units with a bypass system designed to ensure that the rated load current is evenly distributed across the bypass units when the UPS operates in bypass mode, preventing any individual unit from overloading.

The total quantity of single UPS units in a parallel UPS equals $n+r$, where n is the quantity of single UPS units required to support the load; r is the quantity of redundant UPS units.

In a parallel UPS system, one module automatically assumes the role of leader, managing overall control. The other UPS modules are connected to the system and have access to all control parameters to ensure synchronized operation. If any UPS module fails, the load is immediately redistributed among the remaining modules to prevent any disruption. If the lead UPS fails, a different module quickly takes over the leadership role, ensuring continuous system functionality without manual

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intervention.

System Control: A high-speed, redundant, serial communication bus ensures data exchange and communication between the CPUs of each unit. Each module independently controls its function and operational status, coordinating with other modules as needed.

Synchronization: One unit is designated as the reference, and all other units synchronize to it. The reference unit, in turn, synchronizes to the utility bypass voltage if within tolerance. In case of reference failure, another unit automatically assumes the reference role.

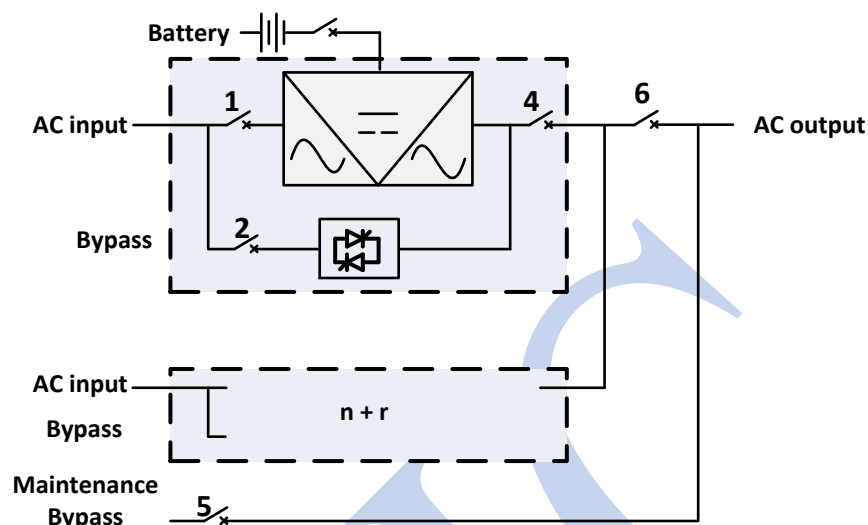
Load Sharing: Inverter output voltage and current are measured on each unit and fed into a load-sharing bus. Any discrepancies between units are automatically equalized.

Setting changes on one UPS are automatically applied to all UPSs in the parallel system.

Required Circuit Breaker/Switches configuration. The figures illustrate the numbering of the breakers. The switches may be outside the UPS unit for high rating.

- Main Input Switch (1).
- Static Switch Input Switch (2).
- Battery Circuit Breaker (3).
- UPS Unit Output Switch (4).
- Maintenance Bypass Switch (5).
- UPS System Isolation Switch (6).

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4.2.3. Modular UPS:

The modular uninterruptible power supply (UPS) system shall consist of multiple units. These units will be connected in a parallel configuration within one or more common frames based on the number of modules required. Each frame shall feature shared input and output buses. Key specifications for the modular UPS include:

- **Maintenance and Static Bypass:** A common maintenance bypass will serve all modules, while the static bypass may either be integrated within each module or centralized, as per manufacturer standards.
- **Battery Configuration:** A shared battery bank with a minimum of two parallel strings shall supply the entire system, with each module featuring its own integrated battery charger. The battery capacity must be expandable by adding independent strings to accommodate increased system power, without requiring system shutdown. A separate battery bank for each module is also acceptable.
- **Hot-Swappability and System Expansion:** The modular system shall support the addition and replacement of modules (hot-pluggable and hot-swappable) without downtime, ensuring seamless scalability. Sufficient space for future module expansion must be provided. All energized work must ensure that hazardous live parts are inaccessible and accessible conductive parts are deenergized and locked out under normal and single-fault conditions, with protective measures implemented to prevent electric burns caused by current flow or arcs, as stated in IEC 60364-4-41.

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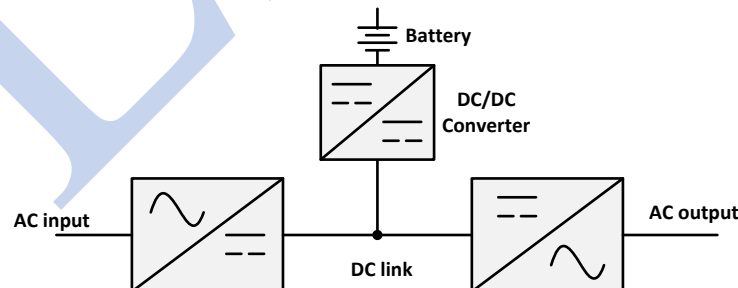
- **Redundancy and Fault Tolerance:** Redundancy, if specified, shall be achieved by adding additional modules or frames, with total capacity. The system must be designed without a single point of failure, ensuring that faults in individual modules do not impact overall functionality.
- **Operational Efficiency and Load Management:** The UPS shall utilize a peer-to-peer architecture for load sharing, synchronization, and selective tripping, without dependence on control interconnections. Automatic load balancing and selective activation of modules based on output load shall optimize efficiency, with unnecessary modules placed in a standby state to reduce energy consumption.
- **Concurrent Maintenance and Serviceability:** Redundant modules must support concurrent maintenance, allowing factory-trained engineers to isolate and service any redundant module while the remaining modules continue to protect the load in normal mode.

4.3.UPS Topology

The UPS system shall be configured with a double-conversion topology, ensuring continuous power supply through a rectifier-inverter combination.

A double-conversion UPS system employs a two-stage power conversion process: AC to DC and DC to AC. In normal operation, the rectifier converts AC input power to DC power, which is then stored in a battery. The inverter converts the DC power back to AC power to supply the load.

When the utility power fails, the UPS seamlessly switches to battery power, ensuring uninterrupted power supply to the load. Once the utility power is restored and meets the specified quality standards, the UPS transitions back to normal operation.



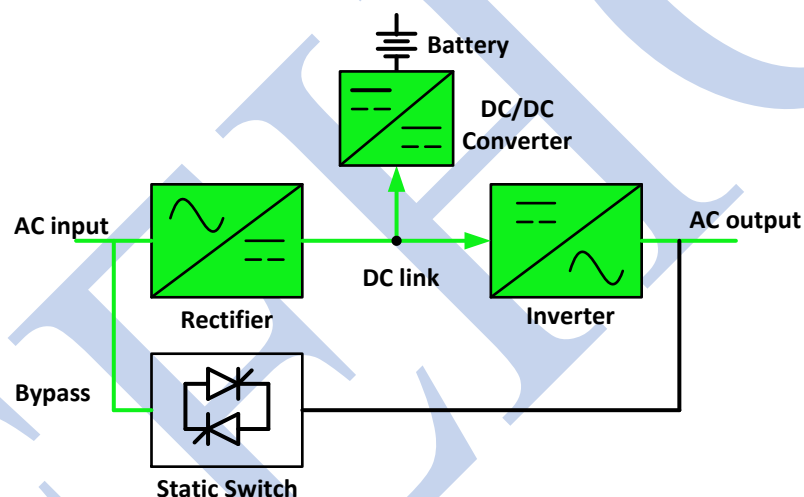
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5. UPS OPERATION MODES

The following specifying the UPS operating modes:

5.1. Normal Mode

Under normal operating conditions, the UPS system ensures continuous power supply to the load. The inverter(s) are synchronized and phase locked with the phase and frequency of the utility grid, enabling a smooth transition between inverter and grid power. The UPS provides conditioned power to the load, maintaining power quality and reliability.

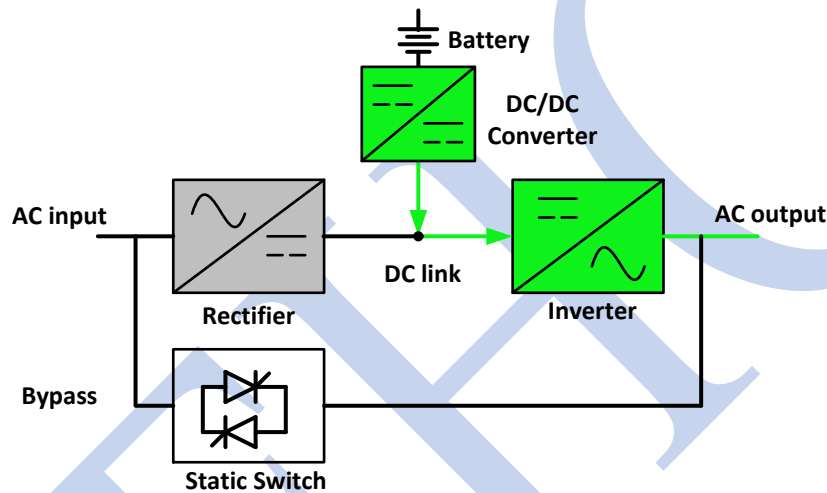


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5.2. Battery Mode

In the event of a utility power outage, the UPS system seamlessly transitions to battery power, continuing to supply conditioned power to the load. The inverters operate on the battery power storage, ensuring uninterrupted power supply.

Once the utility power is restored and meets the specified voltage and frequency criteria, the UPS system automatically re-synchronizes the inverters with the utility grid. This transition is gradual to prevent any disturbances to the load.



5.3. Static Bypass Mode

In this mode, the load is powered directly from the alternative power source. Static bypass modes can be categorized based on the specific conditions and procedures that initiate and terminate bypass operation as follow:

5.3.1. User-initiated Bypass:

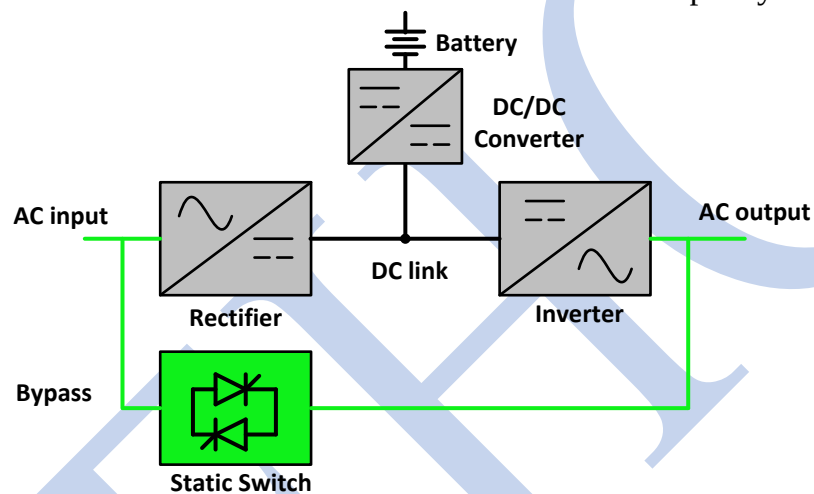
- Initiated by a command by a user from the LCD display or by switch.
- If utility power is interrupted, the system switches to battery mode.

5.3.2. System-initiated Bypass:

- Initiated either by the user turning off the inverter or by an automatic system command in the following cases:
 - Inverter output voltage is outside specified limits.
 - Critical load exceeds the UPS's overload rating.
 - Over-temperature is detected.
 - Malfunction or failure of any UPS module

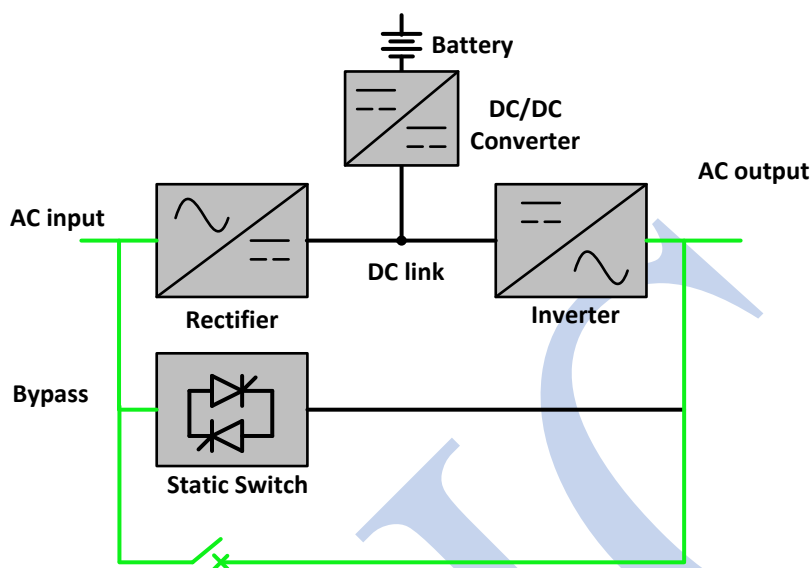
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- Bypass Source Check: Transfer to bypass mode is inhibited, and an alarm is triggered if the bypass power's voltage or frequency are outside acceptable tolerances.
- Automatic Re-transfer to Inverter: The critical load is automatically re-transferred from the bypass source to the inverter (UPS mode) only when the following conditions are met:
 - Inverter output voltage is within specified tolerances.
 - Load current is below the UPS's rated full load capacity.



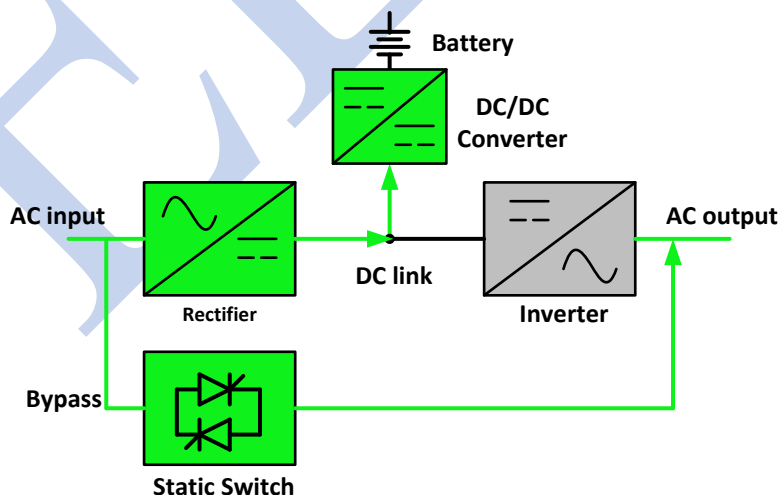
5.4. Maintenance Bypass Mode

The primary objective of this mode is to maintain continuous power delivery to the load by utilizing an alternative power source during maintenance activities. The load is supplied with unconditioned power from the bypass input, and battery power is not available in this mode.



5.5. Economical Mode

Economical Mode-1 primarily benefits by reducing electrical power consumption. This mode configures the UPS to operate on static bypass by default, supplying the load through the bypass path under specific predefined conditions. If a fault is detected, such as bypass voltage or output voltage being out of tolerance, the UPS will immediately switch to normal operation. In case of a utility or mains power interruption, the UPS automatically transfers to inverter operation to ensure uninterrupted power to the load. Battery charging continues while the UPS remains in Economical Mode-1.



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In Economical mode the ups shall achieve the following:

- transition time of less than 6 milliseconds from Economical mode to Normal mode.
- maintain an efficiency of not less than 99%.
- suppress lightning surges while minimizing voltage aberrations.

5.6. General Operation Notes

Upon recovery of the AC input power, the Rectifier will automatically initiate, providing DC power to the Inverter and recharging the Battery. In instances where the Inverter was previously shut down due to low Battery levels, the Load will initially receive power from the Utility via the Automatic Bypass. Once the Battery is sufficiently recharged to support a minimum operational duration under the current Load, the Inverter will automatically restart, and the Load will be transferred back to the Inverter.

The UPS shall be a self-diagnostic type, incorporating a self-test function to ensure proper system operation. This self-test feature shall identify any components of the UPS that may require repair in the event of a fault. Upon system start-up, the UPS will automatically initiate a self-test.

6. UPS ELECTRICAL CHARACTERISTICS

6.1. Ups Input specifications:

- System type: three-phase.
- Nominal Input voltage 400 V.
- Connections 4- wire+ ground.
- Input power factor: > 0.99 from load > 40% to 100 %.
- Total harmonic distortion (THDI): <5% at 100% load <4% at 50% load.
- The input current shall be balanced for unbalanced load conditions ranging from 0% to 100%.
- Adaptive Input Current Ramp-Up No inrush current: The UPS shall gradually adjust its input current over an adaptive period. This gradual ramp-in allows the UPS to manage the power load smoothly, avoiding sudden demand spikes and preventing overloads in the electrical system. Additionally, it mitigates inrush current, the initial surge that occurs when the UPS starts up or reconnects to the AC input power.

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- **Backfeed Protection:** A UPS system must be designed to prevent excessive touch currents from occurring between any input supply terminals during battery operation. This is crucial to mitigate the risk of electric shock.
- To ensure compliance with safety standards, the requirements outlined in IEC 62040-1 for feedback protection must be adhered to. For bypass input and rectifier input.
- **Surge Protection:** The UPS shall incorporate surge protection compliant with IEC 61000-4-5 to mitigate the impact of voltage spikes and surges. For the bypass input and rectifier input.
- The UPS shall remain in normal mode without degradation when connected to an AC input power presenting characteristics as outlined in Table-3:

Table 3: AC input power presenting characteristics

1	Input voltage tolerance	-20 / +15 % of nominal voltage
2	Input frequency tolerance	±10 % of nominal frequency (50 HZ)
3	Voltage unbalance ratio for three-phase type	5%
4	Rated Short circuit withstand current	10 kA for 1.5 sec.
5	Total harmonic distortion of voltage THD _v	≤ 5 % with maximum levels of individual harmonic distortion of voltages not exceeding those in Table-4 in IEC 62010-3
6	Transient voltages, superimposed high-frequency voltages and other electrical noise such as that caused by lightning or capacitive or inductive switching	within category C2 and C3 UPS of electromagnetic immunity levels according to IEC 62040-2 & IEC 61000-4-5

6.2. Ups Output specifications:

- The UPS shall be classified as VFI-SS-11 in accordance with the requirements specified in IEC 62040.
- **Nominal Output Power Rating (kVA):** The UPS shall be rated for [specified kVA value] at a minimum load power factor of 0.9 lagging.
- the rated output voltage tolerance:
 - Balanced Load: ± 1 % @ (0 - 100)% load.
 - Unbalanced Load: ± 2 % @ (0 - 100)% load for 100 % unbalanced load
- voltage unbalance resulting from 100 % load unbalance
 - 50 % Unbalanced Load: ± 2 %.
 - 100 % Unbalanced Load: ± 5 %.

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- Phase displacement tolerance:
 - Balanced Load: $\pm 1\%$ @ (0 - 100)% load.
 - Unbalanced Load: $\pm 3\%$ @ (0 - 100)% load for 100 % unbalanced load
- Output frequency tolerance: Normally synchronized to input line frequency (Bypass) 50 HZ $\pm 0.1\%$. Free running 50 HZ $\pm 0.1\%$.
- Load Power factor: 0.7 lagging to 0.8 leading without derating.
- Overload Capacity:
 - 125% 1 min
 - 150% 10 Sec.
- Voltage transient response for Step Load: $\pm 3\%$ of 100% load step with recovery time of 20 milliseconds to be within 2% of output voltage.
- Total Voltage harmonic Distortion (THDV): $< 5\%$ total harmonic distortion (THD) for 100% Non linear
- output overvoltage protection.

7. UPS OPERATION FUNCTION

The UPS system shall incorporate regulatory and self-protective features to mitigate the following conditions:

- Input overvoltage and undervoltage
- Input power line surges and overcurrent
- Output overvoltage and voltage surges caused by paralleled sources, load switching, or circuit breaker operations
- Sudden output load changes and output short circuit.
- The system shall effectively suppress and eliminate ringing transients, voltage spikes, and surges from the output.
- The UPS system shall achieve an overall efficiency of at least 95% from input to output, when operating at 50% or more of its rated load.
- Temperature sensors shall be installed to monitor the temperature of the UPS system and batteries.
- The UPS system shall be equipped with an emergency shutdown switch. Activating this switch will isolate the UPS system by opening the input, output, and battery contactors, thereby shutting down the rectifier, inverter, and static switch. The emergency shutdown switch shall be protected with a cover to prevent accidental operation as option as perEDC requirements.

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8. EQUIPMENT SPECIFICATION

8.1. Rectifier / Charger Unit

- The Rectifier/Charger Unit may be either a separate unit or integrated into a single unit. However, a separate control system shall be employed for battery charging and discharging.
- The rectifier/charger unit utilizes Active Front End (AFE) technology with Insulated Gate Bipolar Transistors (IGBTs) to convert 3-phase utility voltage into a controlled and regulated DC voltage.
- The rectifier/charger unit shall:
 - Limit Charging Current: Be equipped with input AC current limiting and an adjustable current-limiting feature to control the charging current supplied to the batteries.
 - Regulate Output Voltage: Regulate the output voltage of the charging current.
 - Efficient Recharging: The rectifier/battery charger shall provide adjustable float and fast charging modes for the battery, with voltage, current, and time settings configurable according to the battery manufacturer's recommendations.
 - Sufficient Capacity: Have sufficient capacity to power both the rated load and recharge the batteries simultaneously after a system shutdown.
 - Maintain Battery Charge: Keep the batteries fully charged between discharge cycles.
 - Gradual Power-Up: Limit the initial input current to 20% of the rated output current upon AC power restoration, gradually increasing to 100% over adequate period.

8.2. Inverter Unit

- The inverter converts DC voltage into a three-phase AC voltage, maintaining a constant amplitude and frequency, independent of the AC input voltage.
- The inverter unit shall employ IGBT (Insulated Gate Bipolar Transistor) technology with sinusoidal PWM (Pulse Width Modulation) control.
- Output Voltage Adjustment: The inverter's output voltage shall be adjustable within $\pm 3\%$ of the nominal voltage
- Frequency Regulation: The inverter shall maintain phase-locked synchronization with the bypass source frequency, or, if the bypass frequency is out of tolerance, a built-in temperature-compensated oscillator. The frequency deviation shall not exceed $\pm 0.25\%$.

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- Phase Balance: Electronic controls shall be implemented to regulate each phase, ensuring that unbalanced loading does not cause the output voltage to exceed specified voltage unbalance or phase displacement limits.
- The inverter shall incorporate fault detection and automatic disconnection mechanisms, including a static interrupter and output contactor. Upon fault detection, the inverter control electronics will instantaneously turn off the inverter transistors, while the static transfer switch will simultaneously engage to maintain continuous power to the critical load.

8.3. Static Bypass Switch

- Automatic Transfer: A static bypass switch shall be provided to automatically transfer the load to the bypass source in case of:
 - a system malfunction or overload.
 - Inverter overload capacity exceeded
 - Critical AC load overvoltage or under voltage.
- Seamless Transfer: The transfer shall be executed in a "make-before-break" manner, ensuring uninterrupted power supply.
- Redundancy: The static bypass switch shall be paralleled with a circuit breaker, which can take over if necessary.
- The transfer control logic shall inhibit both manual and automatic transfers of the critical load to the bypass source if any of the following conditions are met:
 - Excessive voltage difference between the inverter and bypass source
 - Bypass frequency exceeding specified limits
 - Bypass and inverter output phase synchronization issues
- Load Return: After an overload-induced transfer to the bypass source, the load can be automatically or manually returned to the inverter once the overload condition is rectified.

9. UPS Batteries

9.1. General

- Battery Type: Sealed Maintenance Free Valve Regulated Lead Acid Battery.
- The batteries shall adhere to the standards outlined in IEC 60896.
- High Discharge Rate Battery
- The battery is designed for a minimum service life of 10 years at 20 or 25°C as per battery manufacturer recommendation, according to the EUROBAT classification.

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The end-of-life is defined as the point when the battery's available capacity drops to 80% of its rated capacity.

- Electrolyte Immobilization System (AGM): This technology uses a fiberglass mat to absorb the electrolyte, preventing it from spilling and ensuring safe operation.
- Low Self-Discharge: They retain their charge for long periods, making them suitable for standby power applications.
- Good Recovery After Deep Discharge: They can withstand deep discharge cycles and recover their capacity.
- Gas Recombination: Over 99% of the gases produced during charging are recombined into water, minimizing gas emissions.
- Low-Pressure Safety Valves: These valves release excess pressure safely, preventing battery damage.
- Flame-Retardant Container: The battery casing is made of flame-retardant material to enhance safety.
- Electrolyte specific gravity shall not exceed 1.25 when fully charged at 25 deg. C.
- Ampere hour rating shall be sufficient for emergency period specified with all inverters operating at full rated output, to a discharge limit of not less than 1.75 V per cell.
- Cells shall be normally maintained at 2.25 V per cell.
- The batteries shall be manufactured within three months prior to delivery to the site.

9.2. Battery Rack Specifications

Mechanical Design

- Robust Construction: The battery rack shall be a sturdy, welded steel frame designed to withstand harsh environmental conditions.
- Adjustable Battery Trays: Equipped with adjustable rails or trays to accommodate various battery configurations and facilitate future upgrades.
- Cable Management: Cables shall be routed along the sides of the rack, away from the battery fronts, for easy maintenance access.
- Corrosion Protection: The rack shall be coated with a durable, corrosion-resistant finish.

Electrical Safety

- Circuit Protection: Each batteries bank shall be protected by a 600VDC circuit breaker for individual fault isolation as per manufacturer recommendations.
- The battery rack shall be equipped with individual temperature sensors for each battery string to enable accurate monitoring and control of the UPS battery system.

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- Grounding: The rack shall be properly grounded to ensure electrical safety.

9.3. Battery Operation System Specifications

- Continuous Battery Monitoring: The system shall continuously monitor battery voltage, current, and temperature to assess overall health and performance.
- The UPS battery monitoring and charging system shall adhere to the manufacturer's recommended charging specifications.
- Predictive Diagnostics: The system shall employ advanced algorithms to predict batteries degradation and estimate remaining lifespan.
- Automatic Battery Testing: Regular automatic tests shall be conducted to verify batteries capacity.
- Intelligent Charging: The system shall optimize the charging process to extend battery life and improve performance taking into account the battery temperature.
- The UPS shall incorporate a battery charging temperature compensation feature to optimize charging performance across various ambient temperature conditions.
- Early Warning System: The system shall provide timely alerts for weak batteries, allowing for proactive maintenance and replacement.
- Deep Discharge Protection: The system shall implement measures to prevent deep discharge, which can significantly reduce battery life.

9.4. Battery Sizing and Selection

The UPS supplier shall perform battery sizing calculations in accordance with IEEE 1184, considering the following parameters:

- Power factor @ (0.9).
- Temperature Factor: A factor of 1.05.
- Aging Factor: A factor of 1.25 to account for battery aging over time.
- Design Margin Factor: A factor of 1.15 to provide a safety margin.
- Cell End Voltage: A minimum cell end voltage of 1.75V.
- Required autonomy: 30 Minutes or 60 Minutes as perEDC requirements.
- Recharge Time: The battery bank shall be capable of recharging to 95% of its capacity within a period recommended by battery manufacturer.

The contractor should provide full batteries specifications, back up time calculations tables based on discharge current and watts per cell with respect to time

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10. Control and Monitoring Functions for the UPS

The UPS system should include an integrated control and monitoring panel, featuring a microprocessor-based unit that provides a clear, operator-friendly interface with a comprehensive status display. This interface should incorporate a single-line mimic diagram, LED indicators, and digital displays that reflect the true layout and status of the system. All controls and monitoring components, such as metering, status, and alarms, should be accessible from the front panel of the UPS cabinet:

10.1. LCD Display and Digital Functions

The system should be equipped with an LCD display (Man-Machine Interface) to enable Full control, measurement reading and historical events, including and not limited to:

- Real-time monitoring
 - Input AC 3-Voltage (VLL, Vph);
 - Input AC 3-ph Current;
 - Output AC 3-Voltage (VLL, Vph);
 - Output AC 3-ph Current;
 - DC Bus Voltage;
 - Input, Output Frequency;
 - Input, output power KW and Percentage;
 - Output Power Factor;
 - Battery charging/discharging current & levels, remaining time,
 - UPS operation Mode
 - Operation of parallel system.
 - Batteries Temperature.
- Alarm monitoring in addition Audible alarms should activate for critical faults
 - Main failure; (audible alarm)
 - Battery discharge;
 - Low Battery;
 - Battery Disconnected (audible alarm)
 - Overloading; (audible alarm)
 - Over/Under of AC I/O Voltage & DC Voltage; (audible alarm)
 - Over temperature. (audible alarm)
 - Bypass out of tolerance. (audible alarm)
 - Any component failure. (audible alarm)
 - Bypass mode operation

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- Control Function
 - Switch on/off Inverter
 - Operation mode control
 - Battery test
 - Alarm reset
- Mimic Diagram: depict a single-line diagram of the UPS system shows the power flow through the UPS and the status of all circuit breakers. Indicator lights should represent power availability, normal operation, and bypass mode, with real-time updates on UPS power paths.
- A history log should store up to 99 events for reference. Diagnostic functions with fault logs displaying the history of events with timestamps.
- Remote Monitoring

The UPS system shall provide voltage-free dry contacts for alarm status and an communication module with interface port via modbus RS 485 or SNMP protocol for workstation or Building Management System connection, enabling remote monitoring of the UPS system's operating status.

11.Additional requirements

- Ventilation
 - Adequate forced-air cooling, provided by appropriately rated ventilation fans, shall be installed to maintain all components within their specified environmental limits.
 - The ventilation fans shall be powered by the UPS system to ensure continuous operation across all UPS operating modes.
 - All ventilation fans and blowers shall be equipped with alarm and remote monitoring capabilities.
- Diesel Generator Interface
 - The Uninterruptible Power Supply (UPS) system must be capable of interfacing with a diesel generator set. The generator set should provide a potential-free contact that closes when the generator starts. Upon receiving this signal, the UPS system should:
 - Optimize Power Consumption: Adjust the rectifier's output voltage to maintain full inverter load without relying on battery charging. This minimizes the overall AC power draw from the generator set.

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- Configure the inverter to operate independently of the incoming AC power frequency
- Protection and Fault Clearing
 - UPS modules must incorporate built-in or inherent electronic current-limiting protection to safeguard against permanent damage, including short-circuit conditions. Rapid-acting, current-limiting devices should protect against failure of solid-state components. Any internal module failure must trigger immediate isolation from input and output sources, either through static interrupters or module shutdown.
 - The UPS system must be protected against, but not limited to, the following:
 - Input overvoltage, undervoltage, and power line surges
 - Output overvoltage and voltage surges
 - Sudden output load changes
 - Output short-circuits
 - Inverter and rectifier overcurrent
 - Auxiliary control circuit overcurrent
- Cable Sizing
 - The cable sizing shall adhere to the guidelines outlined in IEC 60364-5-52 and manufacturer recommendation manual.

12.MARKING:

All the components of the UPS and enclosure should be marked with marking labels according to the applicable IEC standards and should be fitted on all components and enclosure to indicate its technical data. The UPS is marked with mimic diagram to indicate the switch type and its position (i.e.: on / off).

13.DRAWING AND CATALOGUE:

All necessary drawings including layout, dimensions, single line diagram of primary circuits should be submitted with the offer.

Technical catalogues containing the technical data for all offered components of the unit should be submitted in addition to routine maintenance procedure

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14.TYPE AND ROUTINE TESTS:

Table 5 – UPS test schedule

Test description	Routine test	Type test	Subclause
Cable and interconnection check	x	x	6.2.2.2
Light load and functional test	x	x	6.2.2.3
No load	x	x	6.2.2.4
Full load	x	x	6.2.2.5
Synchronization	x	x	6.2.2.6
AC input power failure, VFD test	x	x	6.2.2.7
AC input power return	x	x	6.2.2.8
Transfer to bypass mode	x	x	6.2.2.9
Input – supply compatibility:			
Input voltage tolerance, VI test		x	6.4.1.2
Combined input voltage and frequency tolerance, VFI test		x	6.4.1.3
Rated input current		x	6.4.1.4
Inrush current		x	6.4.1.6
Total harmonic distortion of input current		x	6.4.1.7
Input power factor		x	6.4.1.8
Efficiency		x	6.4.1.9
No load losses		x	6.4.1.10
Standby generator compatibility		x	6.4.1.11
Output – load compatibility:			
Normal mode		x	6.4.2.2
Stored energy mode		x	6.4.2.3
Unbalanced load		x	6.4.2.4
DC voltage off-set		x	6.4.2.5
Load sharing		x	6.4.2.6
Output overvoltage		x	6.4.2.7
Periodic output voltage modulation		x	6.4.2.8
Overload capacity		x	6.4.2.9.1
Fault clearing capacity		x	6.4.2.9.2
Inverter current limit		x	6.4.2.9.3
Dynamic output performance:			
Normal mode to stored energy mode		x	6.4.2.10.2

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Stored energy mode to normal mode		x	6.4.2.10.3
Normal mode to bypass mode – Overload		x	6.4.2.10.4
Step load		x	6.4.2.10.5
Parallel redundant UPS failure		x	6.4.2.10.6
Energy storage device:			
Stored energy time		x	6.4.3.1
Restored energy time		x	6.4.3.2
Battery ripple current		x	6.4.3.3
Restart test		x	6.4.3.4
Charger current limit		x	6.4.3.5
Environmental:			
Shock during transportation		x	6.5.1.2
Free fall during transportation		x	6.5.1.3
Storage in dry heat, damp heat and cold environments		x	6.5.2
Operation in dry heat, damp heat and cold environments		x	6.5.3
Acoustic noise		x	6.5.4
UPS functional unit tests(where not tested as a complete UPS):			
UPS rectifier		x	6.6.2
UPS inverter		x	6.6.3
UPS switch(es)		x	6.6.4
Energy storage device		x	6.6.5

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15. OPTION LIST:

The following table should be filled/attached by ...EDC during tendering.

NO.	Option			
1	UPS Configuration		Single Unit	 KVA
		or	Parallel Unit	(...) n ... KVA+ (...) r ...KVA
		or	modular unit	(...) n ...KVA+ (...) r KVA
2	Shared Batteries bank among multiple UPS units.		☒	☒
3	Emergency shutdown switch		☒	☒
4	Backup period time		30 Minutes	60 Minutes

16.GUARANTEE TABLES:

The tenderer shall fill-in schedules of the technical particulars and performance data.

By signing the tender, the tenderer guarantees that the equipment supplied will confirm to technical and performance data show in such schedules.

Any tender not accompanied with clear and complete guarantee tables should be rejected.

GUARANTEE TABLES FOR UPS

No	Item	Vendor Spcs.
Environmental Conditions		
1	Minimum ambient temperature	°C
2	Maximum ambient temperature	°C
3	Storage temperature	°C
4	Maximum relative humidity	%
5	Protection degree	IP:...
6	Acoustic noise	dB(A)
7	Maximum altitude	m

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System Description		
9	UPS Configuration	
10	UPS Topology	
Ups Electrical Characteristics (UPS Input Specifications)		
1	System type	ph
2	Nominal Input voltage	V
3	Connections Wires	
4	Input power factor	
5	Total harmonic distortion (THD _I):	
6	Adaptive Input Current Ramp-Up No inrush current	(Yes/No)
7	Backfeed Protection	(Yes/No)
8	Surge Protection:	(Yes/No)
	Input voltage tolerance	% of Nominal Voltage
	Input frequency tolerance	% of Nominal Frequency
	Voltage unbalance ratio	%
	Total harmonic distortion of voltage THD _V	%
	Electromagnetic immunity levels	
Ups Electrical Characteristics (UPS Output Specifications)		
1	Nominal Output Power Rating at 0.9 lag power factor	KVA
2	Rated output voltage tolerance: ▪ Balanced Load: @ (0 - 100)% load. ▪ Unbalanced Load:@ (0 - 100)% load for 100 % unbalanced load	
3	Phase displacement tolerance: ▪ Balanced Load: ± 1 % @ (0 - 100)% load. ▪ Unbalanced Load: ± 3 % @ (0 - 100)% load for 100 % unbalanced load.	
4	Output frequency tolerance	
5	Dimensions: Width *Depth *Height	mm*.....mm*.....mm
6	Load Power factor:	lag tolead

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7	Overload Capacity:	☐ % 1 min ☐ % 10 Sec
8	Voltage transient response for Step Load:	 % of 100% load step with recovery time of milliseconds to be within % of output voltage.
9	Total Voltage harmonic Distortion (THD _v):	< % total harmonic distortion (THD) for 100% Non linear
10	Output overvoltage protection.	(Yes/No)
Ups Operation Function		
1	Overall Efficiency	 at 50% or more of its rated load
2	Temperature sensors on UPS and Battery	(Yes/No)
3	Emergency Shutdown Switch	(Yes/No)
Equipment Specification (Rectifier / Charger Unit)		
1	Active Front End (AFE) technology with Insulated Gate Bipolar Transistors (IGBTs)	(Yes/No)
2	Adjustable current-limiting feature	KA
	Regulate Output Voltage feature	(Yes/No)
	Efficient Recharging feature	(Yes/No)
	Sufficient Capacity for supplying battery and rated loads	(Yes/No)
4	Gradual Power-Up: from 20% to 100 %.	(Yes/No)
Equipment Specification (Inverter Unit)		
1	Based on IGBT technology with sinusoidal PWM control	(Yes/No)
2	Output Voltage Adjustment:	 % of nominal voltage
3	Frequency Regulation Adjustment	 % of rated frequency
4	Fault detection and automatic disconnection mechanisms	(Yes/No)

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Equipment Specification (Static Bypass Switch)		
1	Seamless transfer (make-before-break)	(Yes/No)
2	Static Bypass switch accomplish feature in section 8.3	(Yes/No)
UPS Batteries		
1	Battery Type	
2	High Discharge Rate Battery	(Yes/No)
3	Minimum service life	years
4	Low Self-Discharge	(Yes/No)
5	Low-Pressure Safety Valves	(Yes/No)
6	Flame-Retardant Container	(Yes/No)
7	Battery End Voltage	 V
Battery Rack Specifications		
1	Battery rack is accomplished with mechanical design in sec. 9.2	(Yes/No)
2	Battery rack is accomplished with electrical safety in sec. 9.2	(Yes/No)
Battery Operation System Specifications		
1	Battery operation system is accomplished as in sec. 9.3	
Battery Sizing and Selection		
1	Temperature Factor	
2	Aging Factor	
3	Design Margin Factor:	
4	Cell End Voltage	V
5	Autonomy period.	H
Additional requirements		
	Real-time monitoring	(Yes/No)
	Alarm monitoring	(Yes/No)

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	Audible alarms	(Yes/No)
	Control Function	(Yes/No)
	Mimic Diagram	(Yes/No)
	A history log	(Yes/No)
	Remote Monitoring	(Yes/No)
Control and Monitoring Functions for the UPS		
	Ventilation	(Yes/No)
	Diesel Generator Interface	(Yes/No)
	Protection and Fault Clearing	(Yes/No)
	Cable Sizing according to IEC 60364-5-52	(Yes/No)

I / We guarantee the technical data given above for the offered battery charger.

Signature: ----

Date: ---